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MILITARY SPECIFICATION

GENERATOR, ENGINE ACCESSORY: DIRECT-CURRENT 28 VOLT, 25 AMPERE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and inspections of one type of 28 volt (V) direct current (dc), 25 ampere generator which is waterproof and may be direct driven or belt driven.

1.2 Classification. The type covered by this specification shall be one of the generators listed below as specified (see 6.2):

Part Numbers	Item
7355736	Generator, D.C. 28 Volt, 25 Amp
10950808	Generator, D.C. 28 Volt, 25 Amp

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited).
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.
TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-P-636	- Primer Coating, Alkyd, Wood and Ferrous Metal.
TT-P-1757	- Primer Coating, Zinc Chromate, Low Moisture Sensitivity.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.
MIL-C-81562	- Coatings, Cadmium, Tin-Cadmium, and Zinc (Mechanically Deposited).

STANDARDS
FEDERAL

FED-STD-H28/2	- Screw-Thread Standards for Federal Services, Section 2 Unified Inch Screw Threads - UN and UNR Thread Forms.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.

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MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

7355736	- Generator, D.C., 28 Volt (25 Amperes).
10950808	- Generator, D.C., 28 Volt (25 Amperes).

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	- Electrodepostited Coatings of Zinc on Iron and Steel.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish the first completed generator which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent generators delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications and drawings. Materials shall be free of defects and imperfections that might affect the serviceability and appearance of the finished product (see 4.8.1, 4.8.2 and 6.5).

3.2.1 Qualified products. When a part or component is specified to conform to a specification having a Qualified Products List (QPL), the contractor shall maintain documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order, and quantity (see 4.8.1).

3.2.2 Dissimilar metals. Dissimilar metals shall be protected from galvanic corrosion in accordance with requirements of MIL-STD-889 (see 4.8.1).

3.2.3 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials; however, all materials shall be new and unused. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met and that the suppliers' employees are not exposed to undue hazards from the recovered materials (see 6.5).

3.3 Design and construction. Generator assemblies shall conform to Drawing 7355736 or 10950808 and applicable standards and specifications (see 4.8.1 and 4.8.2).

3.3.1 Interchangeability. All generator assemblies and all component parts shall be physically and functionally interchangeable with all other component parts of same part number previously furnished under this specificaton (see 4.8.2).

3.3.2 Standard parts. Military standard parts shall be incorporated wherever applicable. Commercial standard parts may be used, provided they are interchangeable with military standard parts without modification (see 4.8.2).

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3.3.3 Threaded parts. Screw threads of the form, number per inch and class specified on the applicable drawing or military standard shall be in accordance with FED-STD-H28/2 (see 4.8.1 and 4.8.2).

3.3.4 Polarity. Generator shall operate within a negatively grounded electrical circuit (see 4.8.1 and 4.8.2).

3.4 Performance.

3.4.1 Output. The generator shall meet requirements specified on output performance curve of figure 1 (see 4.8.3 and 6.4).

3.4.2 Endurance. The generator shall deliver rated output for 2000 hours. During this time, it shall require no servicing or replacement of parts, except for one change of brushes. The generator shall subsequently conform to 3.4.1 (see 4.8.4).

3.4.3 Overspeed. When operated at 10 000 revolutions per minute (rpm) for 1 minute; the generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings, nor raised commutator bars (see 4.8.5).

3.4.4 Dielectric strength. After application of 500 V root-mean-square (rms) at 60 Hertz (Hz) to armature and field terminals, with the ground brush raised off the commutator and with the field ground terminal disconnected; the generator shall evidence no loosening, cracking, charring, burning or smoking of insulation, or reduction in dielectric strength (see 4.8.6).

3.4.5 Fire hazard. Generator operation shall not cause the explosive mixture of surrounding fuel and air to ignite (see 4.8.7).

3.4.6 Environment.

3.4.6.1 High temperature. During 4 hours of operation at rated output, in 160 ± 5 degrees Fahrenheit (°F) ambient air temperature, the generator frame temperature shall not exceed 320°F. Further, the generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings nor raised commutator bars (see 4.8.8).

3.4.6.1.1 Overload. The generator shall withstand operation at 125 percent (%) of rated output for 10 minutes, applied immediately after operation as specified in 3.4.6.1, and while at that same temperature. The generator shall subsequently conform to 3.4.1 (see 4.8.8.1).

3.4.6.2 Low temperature. After a cold soak for 24 hours, and 4 hours of operation producing rated output at minus 65°F, the generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings, or raised commutator bars (see 4.8.9).

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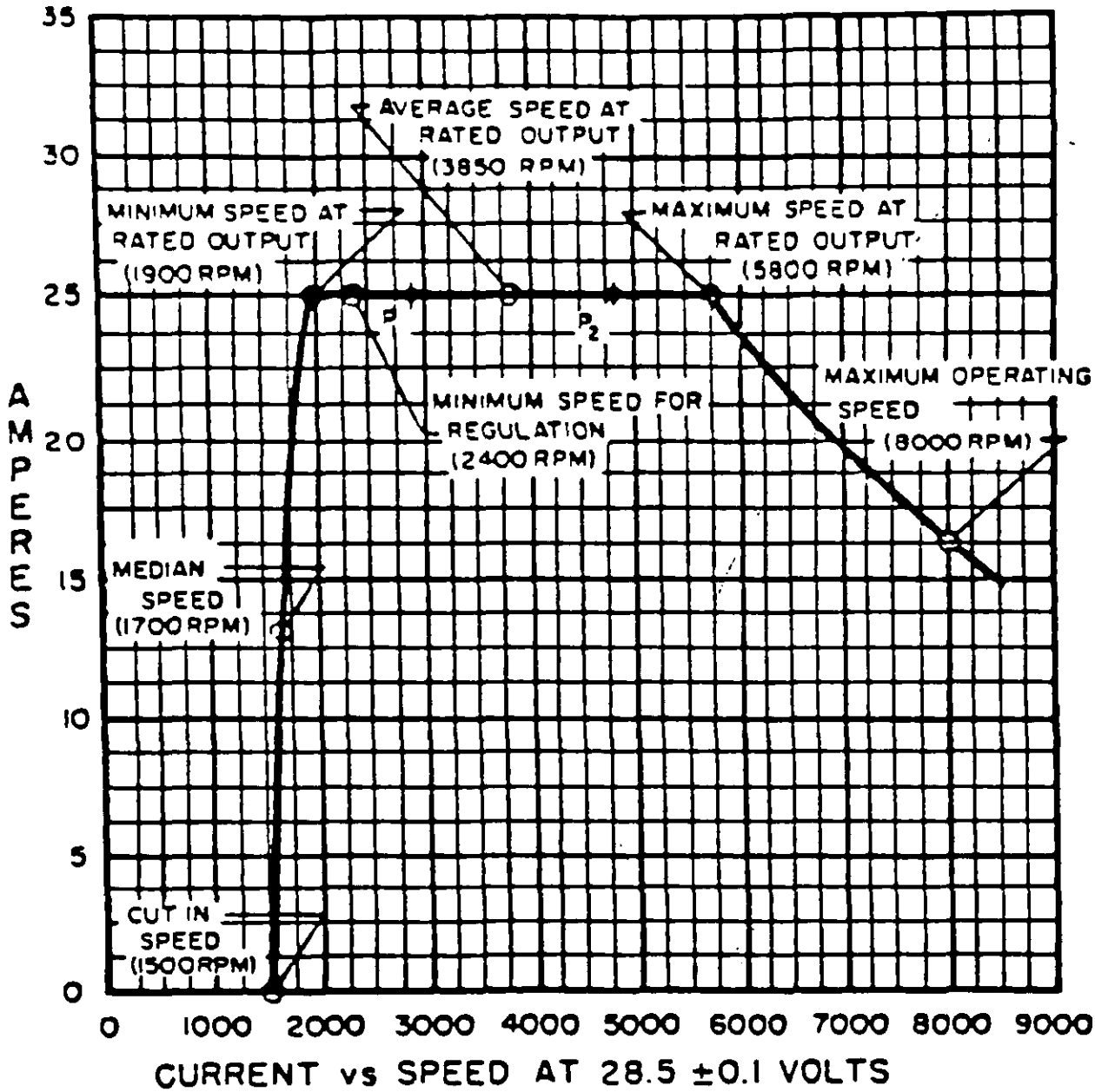


FIGURE 1- TYPICAL OUTPUT PERFORMANCE CURVE

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3.4.6.3 Shock resistance. The generator shall be tested as specified in test condition A, method 213 of MIL-STD-202. The generator shall then be operated as specified in 6.1 and shall conform to 3.4.1 (see 4.8.10).

3.4.6.4 Vibration resistance. The generator shall be tested as specified in method 201 of MIL-STD-202, except the time of test shall be 72 hours, and the generator shall operate as a motor for 1-1/2 of every 2 hours tested. The generator shall then be operated as specified in 6.1 and shall conform to 3.4.1 (see 4.8.11).

3.4.6.5 Waterproofness. The generator interior shall remain dry, during and after total submersion in a salt water solution. The generator shall then operate as specified in 6.1 and shall conform to 3.4.1 (see 4.8.12).

3.4.6.5.1 Reduced waterproofness. For quality conformance testing, the generator shall be tested as specified in procedure 5 of MIL-STD-1184 (see 4.8.12.1).

3.4.6.6 Fungus resistance. The generator shall be tested as specified in method A, class 2 of MIL-F-13927. Before and after testing, the generator shall operate as specified in 6.1 and shall conform to 3.4.1 (see 4.8.13).

3.4.6.7 Corrosion resistance. The generator shall be tested as specified in method 101 of MIL-STD-202, except length of test shall be 200 hours. After testing, the generator shall operate as specified in 6.1 and shall conform to 3.4.1 (see 4.8.14).

3.4.6.8 Sand and dust. The generator shall be subjected to test method 110 of MIL-STD-202, and shall subsequently conform to 3.4.1 (see 4.8.15).

3.4.6.9 Electromagnetic interference. The generator shall conform to electromagnetic interference suppression requirements of MIL-STD-461 for tactical vehicles (see 4.8.16).

3.4.7 Finish. The electrical receptacle assembly shall be cadmium plated. Other exterior surfaces of the generator assembly shall be painted or zinc plated, except for the shaft extension. Screws and other parts shall be painted or zinc plated. No paint shall be applied to mating surfaces on frame or on ends of generator. Cadmium plating shall conform to QQ-P-416, class 2, type II. Zinc plating shall conform to ASTM B633, class 2, type II, or MIL-C-81562, class 2, type II (see 4.8.1 and 4.8.2).

3.4.7.1 Ferrous surfaces. Ferrous metal surfaces shall be cleaned in accordance with the applicable method specified in TT-C-490. Cleaned surfaces shall be treated in accordance with type I or type III of TT-C-490. Exterior surfaces shall be given one coat of primer conforming to TT-P-636, 0.75 to 1.25 mil thick, or TT-P-1757, 0.40 to 0.60 mil thick. Finish coat shall be black gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

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3.4.7.2 Aluminum alloy surfaces. Aluminum surfaces shall be cleaned and treated in accordance with MIL-C-5541, or anodized in accordance with MIL-A-8625. Exterior surfaces shall be given one coat of primer conforming to TT-P-1757, 0.40 to 0.60 mil thick. Finish coat shall be black gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

3.4.7.3 Zinc and cadmium surfaces. When specified (see 6.2), exterior zinc and cadmium surfaces shall be painted with one coat of primer conforming to TT-P-1757, 0.40 to 0.60 mil thick. Finish coat shall be black gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

3.5 Identification and marking (see 4.8.1 and 4.8.2).

3.5.1 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.5.2 Identification. A nameplate shall be affixed to the exterior of the generator frame in a location specified on applicable drawings or, if not specified, in a conspicuous place. Nameplates shall conform to MIL-P-514 and shall include the following minimum information:

Generator, 28 Volts DC.
Amperes, _____.
Military part number.
National stock number.
Manufacturer's serial number.
Manufacturer's identification.
Manufacturer's part number.
US

3.5.3 Permanent mark. When specified (see 6.2), the military part number required on nameplates shall be permanently inscribed on the generator frame in such location as to be covered by the nameplate.

3.6 Workmanship. Workmanship shall be such as to provide a generator free from defects such as: cracked, broken, loose, or dented parts or components; scratches, defective paint, rust burrs and sharp edges. All traces of corrosive flux shall be neutralized or removed after soldering (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 % of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662. Measuring instruments shall be accurate within tolerances specified in table I.

TABLE I. Measuring instrument tolerances.

Measuring instrument	Tolerance
Voltmeter	.25 % of full scale
Ammeter DC	.25 % of full scale
Tachometer	1.00 % of full scale
Temperature indicating system	± 5°F

4.2 Classification of inspections:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 23 ± 10°C
b. Barometric pressure 725 + 50 millimeter (mm) mercury (Hg)
   - 75
c. Relative humidity 50 ± 30 %

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4.3.1 Climate for testing. Tests shall be conducted in a laboratory free of dust, drafts, fumes and products of combustion. Samples shall be frame temperature stabilized for 1 hour before being subjected to tests (see 4.3.3 and 4.3.4). The applicable regulator shall be connected in the test circuit for all generator test speeds.

4.3.2 Apparatus. In addition to standard measuring equipment (see table I, and applicable test specifications), test apparatus shall include chambers for conducting low temperature (minus 80°F), high temperature (plus 200°F), sand and dust tests, and a submersion tank.

4.3.3 Frame temperature. The frame temperature shall be measured by means of a thermocouple attached to the pole screw furthest from the source of cooling air.

4.3.4 Stabilized frame temperature. The generator system shall be considered to be frame temperature stabilized when operation under constant conditions causes a change in frame temperature of not more than 1°F in any 15 minute period.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select 3 generators produced under the production contract for first article inspection. First article samples shall be examined as specified in 4.5.2 and tested as specified in tables II and III. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply generators that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

TABLE II. Order of first article testing.

Specimens		Test
1	4.8.3	Output
	4.8.5	Overspeed
	4.8.8	High temperature
	4.8.8.1	Overload
	4.8.9	Low temperature
	4.8.11	Vibration
	4.8.14	Corrosion resistance
	4.8.7	Fire hazard
	4.8.15	Sand and dust
	4.8.12	Waterproofness
	4.8.6	Dielectric strength
	4.8.16	Electromagnetic interference

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TABLE II. Order of first article testing - Continued.

Specimens		Test
2	4.8.3	Output
	4.8.5	Overspeed
	4.8.8	High temperature
	4.8.8.1	Overload
	4.8.9	Low temperature
	4.8.10	Shock resistance
	4.8.13	Fungus resistance
	4.8.6	Dielectric strength
3	4.8.3	Output
	4.8.4	Endurance

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article test shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Tests	Control
Materials and construction	3.2 thru 3.2.2, 3.3.3, 3.3.4, 3.4.7 thru 3.5.3	4.8.1	X			
Defects (see 4.5.2 and table IV)	3.2, 3.3 thru 3.3.4, and 3.4.7 thru 3.6	4.8.2	X	X		
Output	3.4.1	4.8.3	X		X	X
Endurance	3.4.2	4.8.4	X			
Overspeed	3.4.3	4.8.5	X			X

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TABLE III. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance Examination	Quality conformance Tests	Control
Dielectric strength	3.4.4	4.8.6	X		X	
Fire hazard	3.4.5	4.8.7	X			
High temperature	3.4.6.1	4.8.8	X			
Overload	3.4.6.1.1	4.8.8.1	X			
Low temperature	3.4.6.2	4.8.9	X			
Shock resistance	3.4.6.3	4.8.10	X			
Vibration resistance	3.4.6.4	4.8.11	X			
Waterproofness	3.4.6.5	4.8.12	X			X
Reduced waterproofness	3.4.6.5.1	4.8.12.1			X	
Fungus resistance	3.4.6.6	4.8.13	X			
Corrosion resistance	3.4.6.7	4.8.14	X			
Sand and dust	3.4.6.8	4.8.15	X			
Electromagnetic interference	3.4.6.9	4.8.16	X			

4.5 Quality conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the generators of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.1.3 Sampling for acceptance tests. Samples for acceptance tests shall be selected in accordance with level S-3 of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

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Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table IV.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Nonconformance in design and construction (see 3.2 and 3.3.4).	Visual and gage
102	Incorrect dimensions affecting interchangeability (see 3.3.1 and 3.3.2).	Gage
103	Faulty workmanship affecting performance (see 3.6).	Visual
Minor	AQL 2.5% Defective	
201	Incorrect dimensions not affecting interchangeability (see 3.2 and 3.3.3).	Gage
202	Improper finish (see 3.4.7).	Visual
203	Improper marking (see 3.5).	Visual
204	Faulty workmanship affecting appearance (see 3.6).	Visual

4.5.3 Tests. Samples selected in accordance with 4.5.1.3 shall be subjected to the quality conformance tests specified in table III, in the order specified in table V and shall conform to an AQL of 6.5.

TABLE V. Order of quality conformance tests.

Requirement		Test
3.4.4	Dielectric strength	4.8.6
3.4.1	Output	4.8.3
3.4.6.5.1	Reduced waterproofness	4.8.12.1

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4.6 Control tests. Control tests shall be conducted on 1 generator from each lot of 500 units consecutively produced, except that not more than three tests shall be performed in a 1 month period, nor less than one test in a 1 month period. The generators shall be identified as to production period and from a production lot which has passed the examination specified in 4.5.2.2 and quality conformance tests specified in 4.5.3. Control tests shall be as specified in table III and conducted in the order specified in table VI.

TABLE VI. Control tests.

Requirement		Test
3.4.1	Output	4.8.3
3.4.6.5	Waterproofness	4.8.12
3.4.3	Overspeed	4.8.5

4.7 Failure. Failure of any generator to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.2.2, 3.3.3, 3.3.4, 3.4.7 through 3.5.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data.

4.8.2 Defects. Conformance to 3.2, 3.3 through 3.3.4, 3.4.7 through 3.6 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Output. To determine conformance to 3.4.1, the generator shall be stabilized at the highest attainable frame temperature, not exceeding 180°F above ambient temperature, at 28.5 V. No less than six points shall be measured on the performance curve, including the following:

a. Cut-in speed (see 6.4.1). This speed shall be determined with no resistance in series with the field.
b. Minimum speed at rated output (see 6.4.2). This speed shall be determined with no resistance in series with the field.
c. Maximum speed at rated output (see 6.4.4). This speed shall be established by adjusting field resistance.
d. Maximum operating speed (see 6.4.7). This speed shall be established by adjusting field resistance.

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4.8.4 Endurance. To determine conformance to 3.4.2, the generator shall be operated as specified in 6.1, with voltage output regulated to 28.5 ± 0.1 V dc according to the following schedule of times and shaft speeds:

	Time	Speed
	333 1/3 hours	2400 rpm
	333 1/3 hours	5800 rpm
	333 1/3 hours	3850 rpm
1/	As required	As required
	333 1/3 hours	2400 rpm
	333 1/3 hours	5800 rpm
	133 1/3 hours	3850 rpm
2/	100 hours	3850 rpm
3/	100 hours	3850 rpm

1/ At this time, generator shall undergo test of waterproofness as specified in 4.8.12.
2/ Generator positioned so armature shaft forms angle of 45° ± 1° with horizontal.
3/ Generator position adjusted to elevate opposite end (see 2/) of armature shaft until shaft forms angle of 45° ± 1° with horizontal.

Output current, voltage, and generator temperature shall be measured periodically during this procedure.

4.8.5 Overspeed. To determine conformance to 3.4.3, generator frame temperature shall be stabilized (see 4.3.3) at maximum operating speed (see 6.4.7). Field circuit shall then be opened and generator speed increased to 10 000 rpm for 1 minute.

4.8.6 Dielectric strength. To determine conformance to 3.4.4, the generator shall be operated as specified for 1 minute. Potential shall be applied to the armature and field terminals with the ground brush raised off the commutator, and with the field ground terminal disconnected. (The test may be conducted on frame and field, and armature subassemblies prior to generator assembly.) The generator shall then be examined.

4.8.7 Fire hazard. To determine conformance to 3.4.5, the generator shall be subjected to the following conditions for hazardous sparking inspection.

4.8.7.1 Apparatus. Apparatus for conducting this test shall include a chamber and power sources required for operation of the chamber and generator. The chamber shall include following features:

a. A pressure release system which operates when an explosion occurs within the chamber. The system shall be so designed that danger to operating personnel is minimized.
b. A hinged or removable, gasket-sealed door of sufficient size to facilitate installation of the generator within the chamber.

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c. A means of obtaining adequate circulation of vapors within the chamber.
d. A window for safe observance of equipment under test.
e. Cable ports, for insertion of needed cables, designed to facilitate a pressure-tight seal around cables.
f. Openings for installation of pressure-sealed remote controls.
g. Devices to indicate temperature within the chamber.

4.8.7.2 Procedure. The generator and necessary operating controls shall be installed in the chamber, operated to determine that they are functioning properly, and observed to locate any sparking which might be a potential hazard. The chamber shall be sealed, and a predetermined quantity of gasoline conforming to MIL-G-3056 shall be vaporized and drawn into chamber. The quantity of gasoline used shall result in a concentration of fuel-to-air within limits of 3-1/2 to 5 % by weight. The Generator shall be operated at maximum speed at rated output for 10 minutes to determine conformance to 3.4.5.

4.8.8 High temperature. To determine conformance to 3.4.6.1, the generator shall be stored for 24 hours in 160 ± 5°F ambient air temperature. Frame temperature (see 4.3.3) shall be stabilized (see 4.3.4) by operating at minimum speed for regulation (see 6.4.6), at rated output, in the same ambient air temperature. When the frame temperature is stabilized, the generator shall be operated at rated output for 4 hours in the same ambient temperature. The generator shall be examined and frame temperature recorded.

4.8.8.1 Overload. To determine conformance to 3.4.6.1.1, the generator, after successfully completing the high temperature test, shall be operated as specified and then shall meet the requirements of 3.4.1.

4.8.9 Low temperature. To determine conformance to 3.4.6.2, the generator shall be stored for 24 hours and shall be frame temperature stabilized (see 4.3.4) in minus 65 ± 5°F ambient air temperature. Stabilization shall be attained by operating at minimum speed for regulation (see 6.4.6). Thereafter, the generator shall be operated at rated output for 4 hours and subsequently examined.

4.8.10 Shock resistance. To determine conformance to 3.4.6.3, the generator shall be tested as specified and then shall be inspected to meet the requirements of 3.4.1.

4.8.11 Vibration resistance. To determine conformance to 3.4.6.4, the generator shall be tested as specified, except that the duration of vibration shall be 24 hours in each direction of each of the 3 major axes (total of 72 hours). For 1-1/2 hours of every 2 hours in each plane, the generator shall be operated as a motor at a speed of 4500 ± 500 rpm. Subsequently, the generator shall meet the requirements of 3.4.1.

4.8.12 Waterproofness. To determine conformance to 3.4.6.5, the generator shall be operated and inspected as specified.

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4.8.12.1 Reduced waterproofness. To determine conformance to 3.4.6.5.1, the generator will be tested as specified for quality conformance inspection.

4.8.13 Fungus resistance. To determine conformance to 3.4.6.6, the generator shall be tested as specified except that the test shall be accomplished without disassembly, and the period of exposure shall be a continuous 90 days. The generator shall then be operated for 1 hour at 6500 ± 500 rpm without load, and shall subsequently meet the requirements of 3.4.1.

4.8.14 Corrosion resistance. To determine conformance to 3.4.6.7, generator shall be tested as specified except that length of test shall be 200 hours. The generator shall then be operated for 1 hour at 6500 ± 500 rpm without load, and shall subsequently meet the requirements of 3.4.1.

4.8.15 Sand and dust. To determine conformance to 3.4.6.8, the generator shall be tested as specified, except that during 10 minutes of each hour, the generator shall be operated as a motor at a speed of 4500 ± 500 rpm. After testing, the generator shall meet the requirements of 3.4.1.

4.8.16 Electromagnetic interference. To determine conformance to 3.4.6.9, generator shall be tested as specified in applicable provisions of MIL-STD-462.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. Generators covered by this specification are intended for use in military motor vehicles, and on other military equipment, to produce direct current electrical energy for requirements such as lighting and battery-charging.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Generator drawing number (see 1.2).
c. If first article samples are required (see 3.1).
d. If exterior zinc and cadmium surfaces are painted (see 3.4.7.3).
e. If permanent marking shall be inscribed on frame (see 3.5.3).
f. If responsibility for inspection shall be other than specified (see 4.1).

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g. If responsibility for inspection equipment shall be other than specified (see 4.1.2).
h. If inspection conditions are other than specified (see 4.3).
i. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Subject term (key word) listing.

Accessory, engine generator, 28 volt, 25 ampere.
DC generator, 28 volt, 25 ampere.
Engine accessory, DC generator, 28 volt, 25 ampere.
Engine generator, accessory, 28 volt, 25 ampere.

6.4 Definitions.

6.4.1 Cut-in speed. Cut-in speed is the lowest speed at which the generator, with frame temperature stabilized, will generate 28.5 V with no external load.

6.4.2 Minimum speed at rated output. Minimum speed at rated output is the lowest speed at which rated output can be obtained at 28.5 V.

6.4.3 Median speed. Median speed is the lowest speed at which 50 percent of rated output can be obtained at 28.5 V.

6.4.4 Maximum speed at rated output. Maximum speed at rated output is highest speed at which rated output is generated at 28.5 V.

6.4.5 Average speed at rated output. Average speed at rated output is speed half way between "minimum speed at rated output" and "maximum speed at rated output".

6.4.6 Minimum speed for regulation. Minimum speed for regulation is minimum speed at rated output plus 500 rpm.

6.4.7 Maximum operating speed. Maximum operating speed represents maximum speed at which generator may be expected to perform in normal operation.

6.5 Recycled materials.

6.5.1 Recovered materials. 'Recovered materials' means materials that have been collected or recovered from solid waste (see 6.5.2) to become a source of new materials, as distinguished from virgin raw materials.

6.5.2 Solid waste. For definition of 'solid waste', refer to section 23.402 of the Federal Acquisition Regulations.

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6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - AS
Air Force - 99

Review activity:
DLA - CS

User activity:
Navy - MC

Preparing activity:
Army - AT

(Project 2920-0417)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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